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AIM :

To find the top 5 best selling products of the past month.

Software Used :

Python 3, Pandas (Jupyter notebook)

Algorithm :

1. Create the sales DataFrame (product, quantity, date).
2. Filter the rows from the past month.
3. Group by product and sum the quantity.
4. Sort in descending order and take the top 5.
5. Display the result.

CODE :

```
import pandas as pd
sales = pd.DataFrame({
    "product": ["Pen", "Book", "Bag", "Pen", "Bottle",
                "Book", "Lamp"]
    "quantity": [10, 4, 2, 8, 6, 5, 3, 12, 4, 1]
    "date" : pd.to_datetime(["2024-05-02", "2024-05-04",
                             "2024-05-06", "2024-05-09",
                             "2024-05-20", "2024-05-22"].

3)
month = sales[(sales["date"] >= "2024-05-01") &
                        (sales["date"] <= "2024-05-
                                            31")]
print(month.groupby("product")["quantity"].sum().
                                        nlargest(5))
```

Input/Output :

Pen 35 | Book 15 | Bottle 13 | mouse 12 | Bag 5 .

Result :

The top 5 products are Pen, Book, Bottle ; mouse & Bag. Pen is the best seller with 35 units.